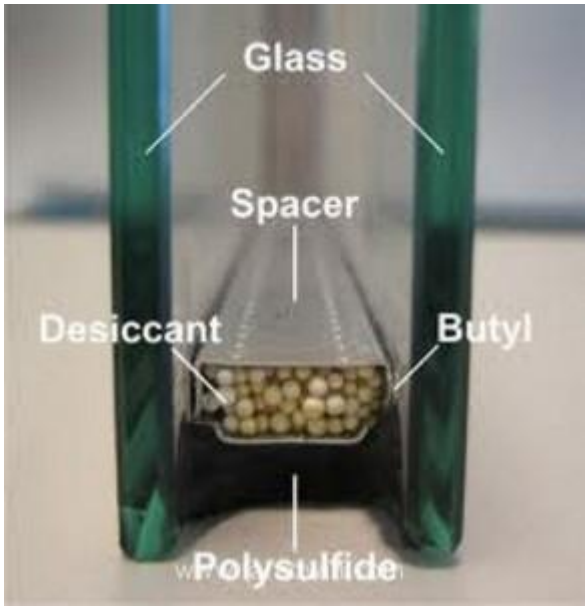
Designing a Mullion Cutting Machine using CATIA software

Abstract:

Windows bring light, beauty and warmth to the house and are sources for our awareness of the change of time and familiarity with the local situation. But in addition to creating a beautiful view, they cause 40% energy loss, which can be easily solved with new technology in the production of glasses.

Double-glazed windows are made up of double-glazed glass plus the profile that covers it, the window profiles in the world markets have different products, the most common of which are PVC, galvanized and plum, which European countries are moving towards PVC windows.





Due to that fact, we should build windows with the least amount of that waste and for this we need to work with modern industry machines. Mullion catting machine is one of those machine which works with UPVC windows and doors.

What is UPVC?

UPVC stands for Unplas PolyVinyl Chloride. The main component of UPVC profile is (PolyVinyl Chloride).

This material makes up about 85 percent of the primary composition of the UPVC profile production.

In addition to this substance, other additives are also added to the composition to create the required properties, the absence of which or the change in the amount used in the formulation can severely affect the final product.

Additives can be mentioned as follows:

1- Fillers Fillers

2- Lubricants

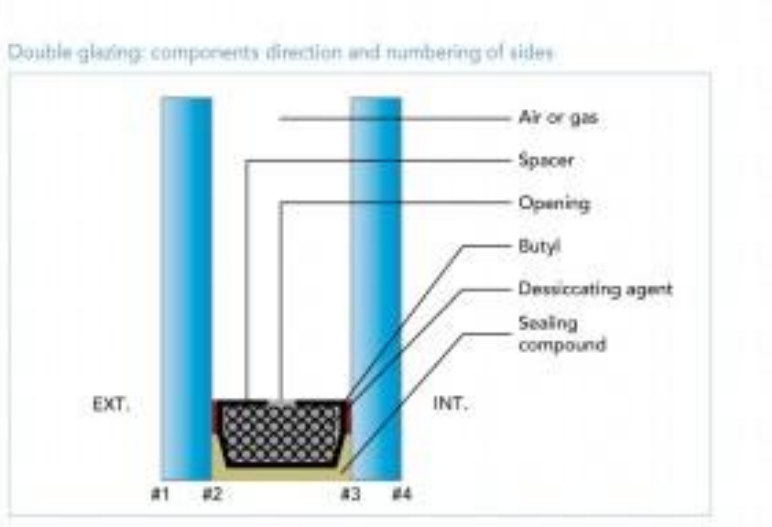
3- Stabilizers thermal stabilizers

4- Color stabilizers

5- Helping processes

6- Impact modifiers against impact

Air and gas are weaker thermal conductors. The air or gas between the two panes of glass causes a lack of heat exchange between the outside and the inside, making your residence cool in summer and warm in winter. The use of double-glazed glass has reduced energy loss by conduction method, and due to the distance between the two panes, the amount of energy transfer by displacement method is extremely small.



The difference between the insulating properties of single-pane and double-pane glass is very significant.

Parallel to the increase in the price of energy, everyone is trying to save on its consumption and naturally reduce the amount of energy carriers' bills. The use of double-glazed windows leads to a significant reduction in energy costs. Despite the higher price of double-glazed glass compared to single-glazed glass, this price difference will be compensated in a short period of time due to the savings in energy consumption costs.

The main features of double-glazed glass:

Reduce energy loss

Preventing the formation of steam between the double walls

More safety than single glass windows

production technology:

In the structure of double-glazed glass, two layers of glass are separated by a layer of aluminum profile.

The above structure is integrated into an integrated set by means of special adhesives and placed inside the window frame. Moisture-absorbing material is placed inside the aluminum profile so that the interior is completely dry. The inner space between two glasses is filled with dry air or an inert gas (such as argon).

In a window, choosing the type of glass is very important. Nowadays, with the advancement of technology, the windows have become more beautiful and according to various special and practical properties

In the building, glasses have been produced that have minimized energy loss.

Windows transmit energy in four ways:

Heat transfer to a colder surface (Conduction)

Heat transfer to the surrounding air (Convection)

its reflection (Radiation)

Constant transfer of heat from the seam and tires next to the window (Air Leakage)

Today, with the technology used in the production of double-glazed doors and windows and due to special layering and the use of double-glazed glass, the amount of light, heat and sound entering the living environment is controlled and the fading of carpets, carpets and home furniture is prevented.

The invention of double-glazed glass has undoubtedly been one of the most important developments in the construction industry. Better use of sunlight and at the same time full control of light, heat, cold, sound are the main advantages of this type of glass. By preventing heat exchange, double-glazed glass creates significant savings in the purchase costs of building heating and cooling facilities.

In the long term, double-glazed windows also reduce the energy and management costs of the building. In cold weather, perspiration on the surface of the glass inside the building, in addition to reducing visibility, causes serious damage to the building materials around the windows and causes a decrease The humidity inside the building is also reduced.

For this reason, in modern architecture, using insulated windows and installing double-glazed windows on it has become a common thing.

Required machines:

Some of the machines needed in this industry are:

Mullion device

Single head cutting machine

Double cutting machine

Teksar welding machine

Double welding machine

Milling copy machine

Zawar cutting machine

Mullion Cutting device:

Cutting and milling the ends of UPVC middle profiles

Ease of milling UPVC profiles with different angles

• The system of taking UPVC parts pneumatically

• Single axis for a set of cutting blades

• Fast and easy operation due to the possibility of changing blades easily

The possibility of using two different types of UPVC profiles at the same time

- Application in cutting UPVC and aluminum profiles

- The possibility of cutting at angles between 45 and 90 degrees

- Working with a fully automatic pneumatic system

- Ability to adjust the speed of rotation of the saw and the speed of cutting

- Full security of work when cutting aluminum parts with automatic caps system and downward rotation of the saw

- Has a movable control panel

- The possibility of adding an automatic conveyor system if requested by the customer

Device features

Machine length: 750 mm, machine width: 920 mm

Machine height: 1160 mm, machine weight: 205 kg

Saw diameter: 450 mm, wind pressure: 6-8 bar

AC motor: 2 HP, 3-phase, 3000 rpm Voltage: 3-phase, 380 V, 50 Hz

Total power consumption of the device: 2200 watts



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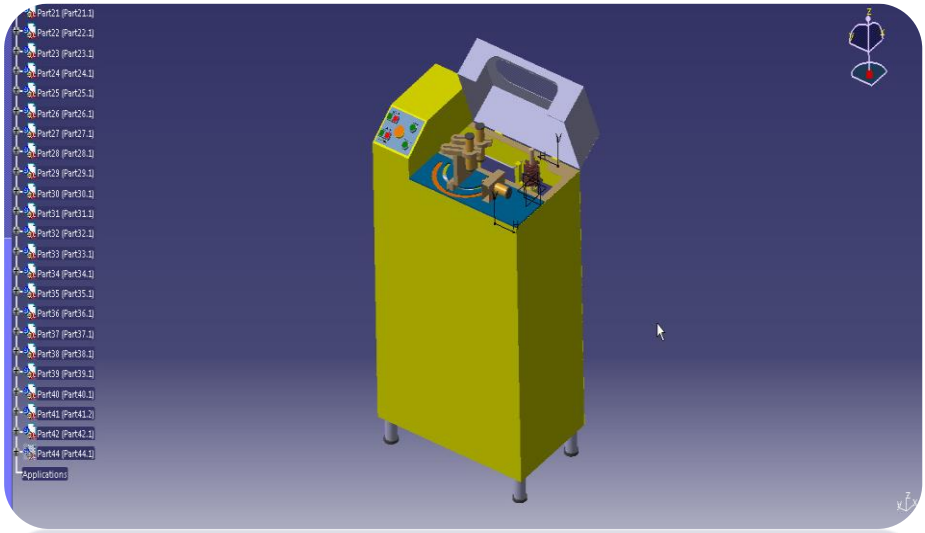
This 3D design has been done by the powerful CATIA software and reverse engineering, so that all the sizes are taken from the machine available in the company for the production of double-glazed doors and windows, and then it is designed with the software.

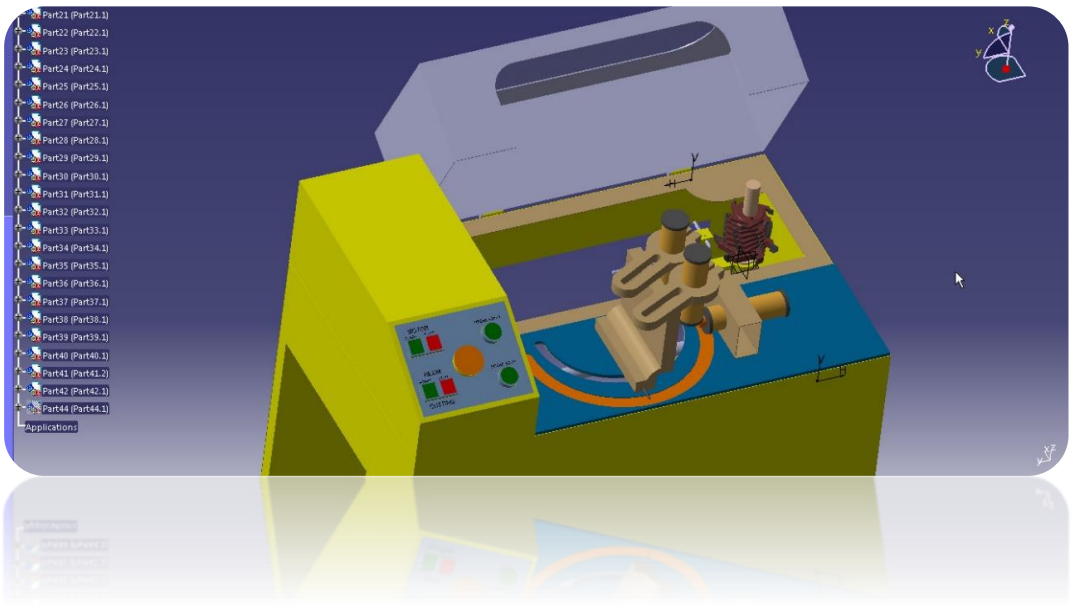
Original machine figure:

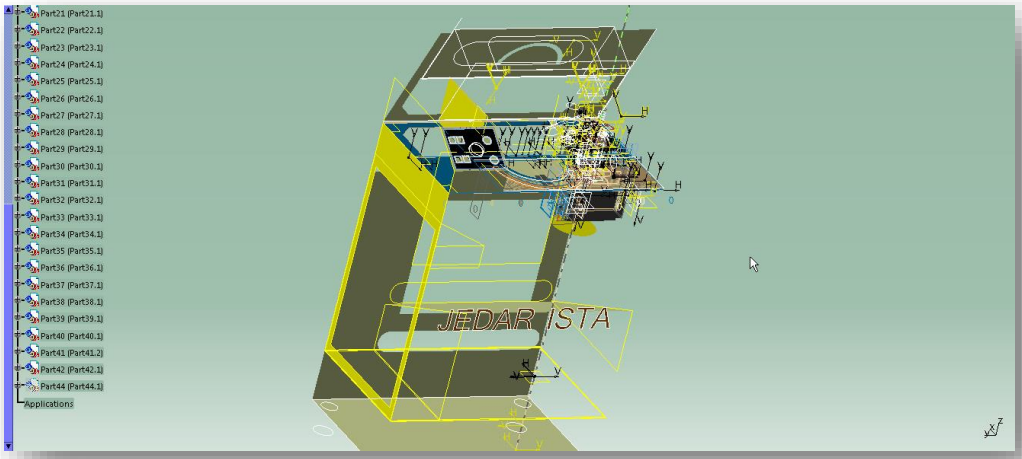


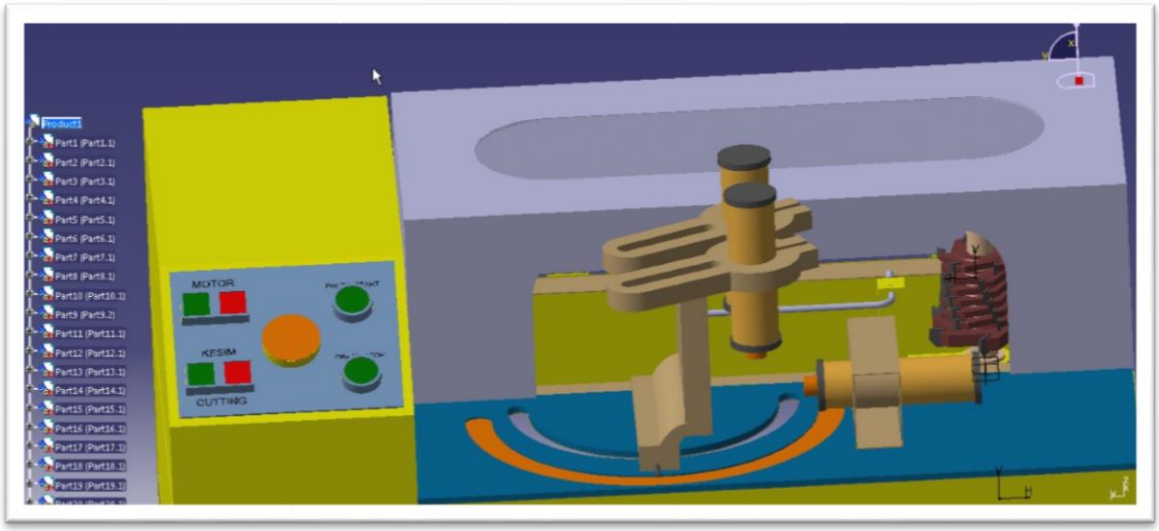


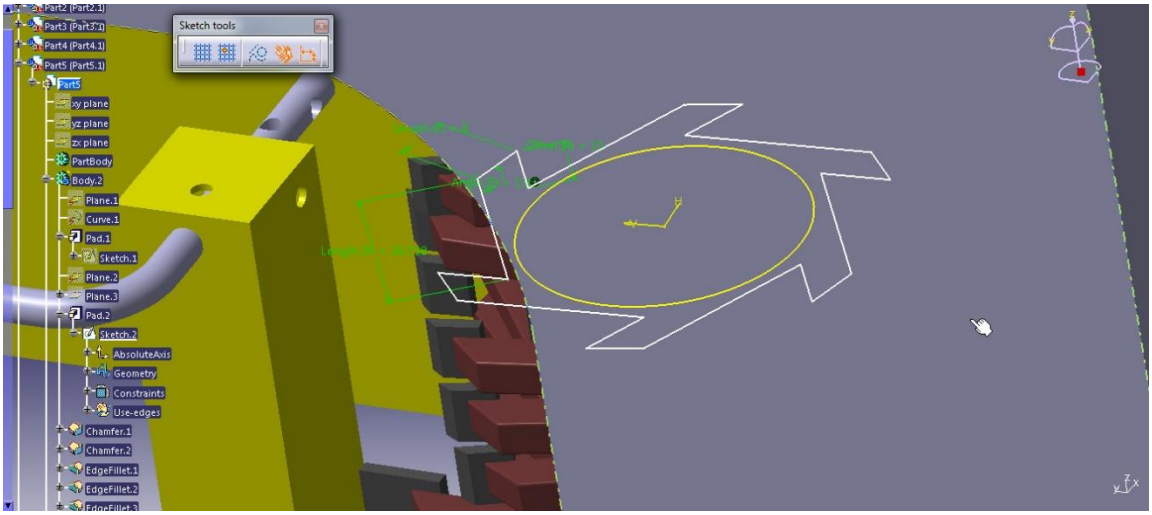
Application Environment:



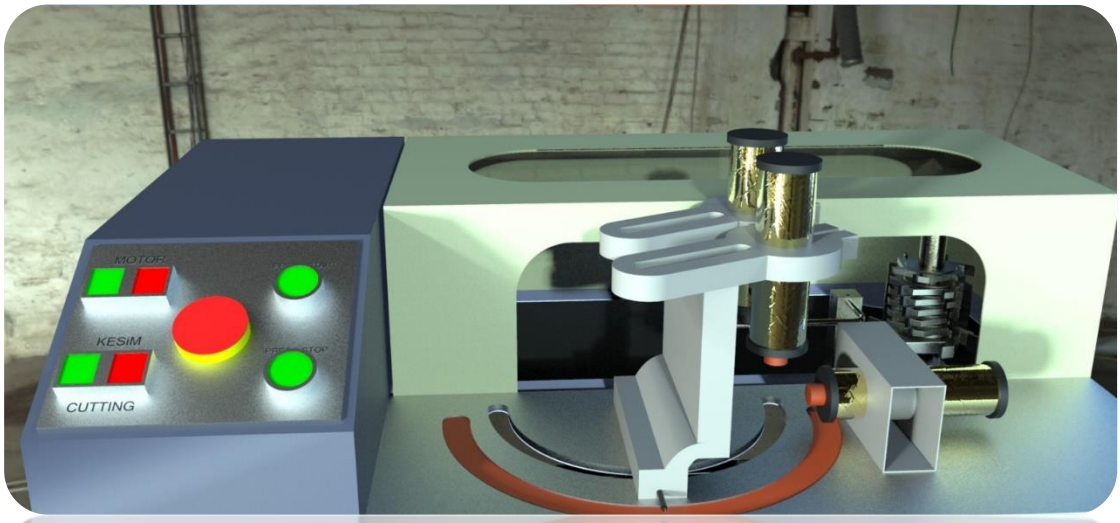


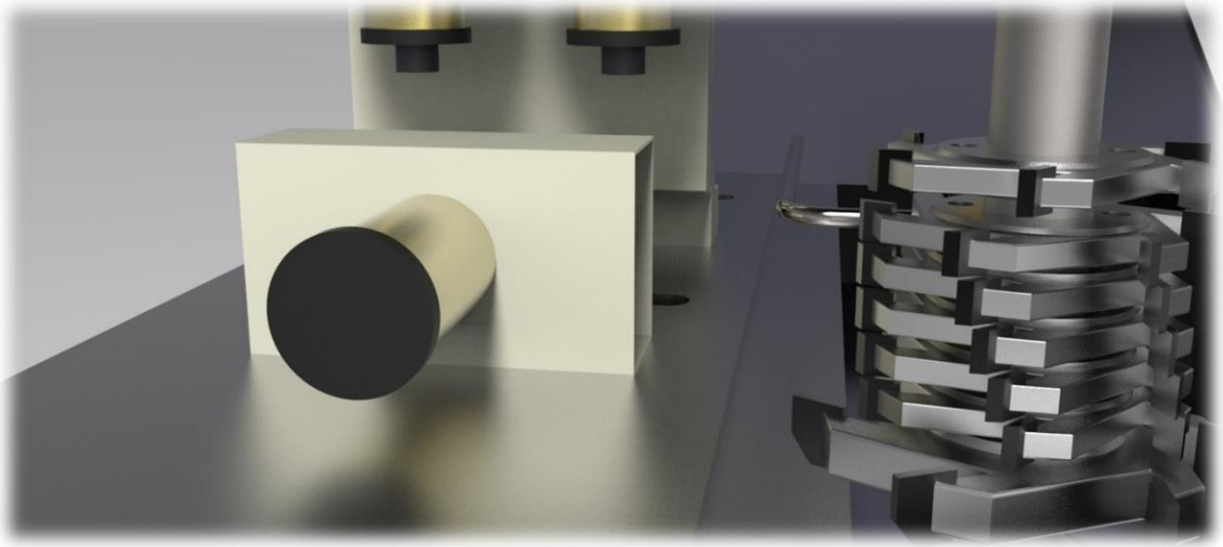

JEDARISTA


MOTOR
KESIM
CUTTING

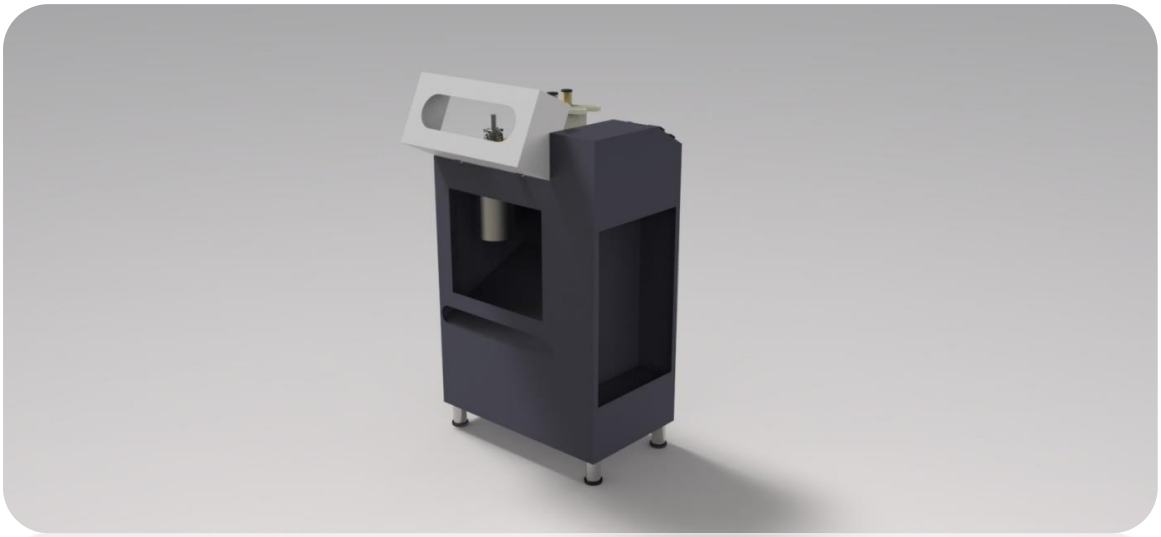

Sketch tools

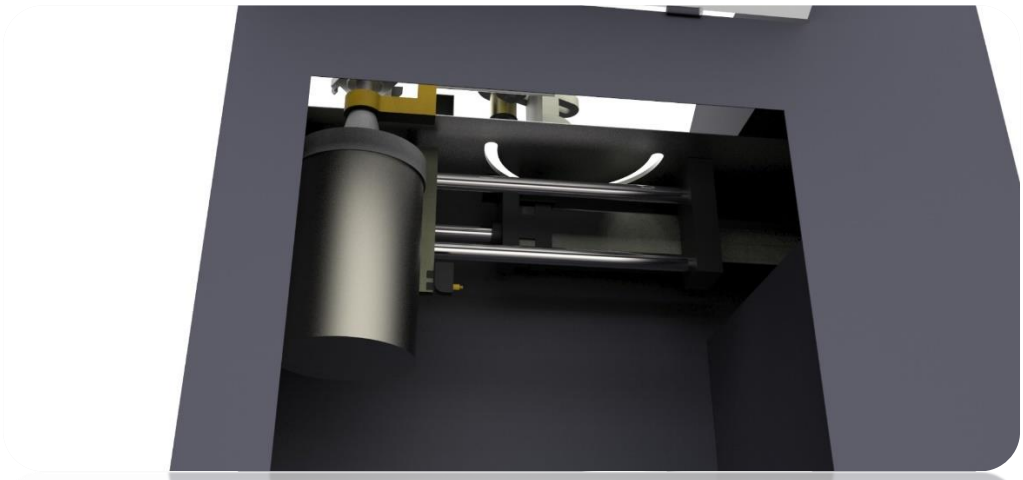
Final Renderings:












MOTOR
START
STOP
PRESS START
KESIM
START
STOP
PRESS STOP
CUTTING